These are grand claims and, if correct, represent a monumental step forward. But the most stunning feature of superstring theory, one that I have little doubt would have set Einstein's heart aflutter, is its profound impact on our understanding of the fabric of the cosmos. As we will see, superstring theory's proposed fusion of general relativity and quantum mechanics is mathematically sensible only if we subject our conception of spacetime to yet another upheaval. Instead of the three spatial dimensions and one time dimension of common experience, superstring theory requires *nine* spatial dimensions and one time dimension. And, in a more robust incarnation of superstring theory known as *M-theory*, unification requires *ten* space dimensions and one time dimension—a cosmic substrate composed of a total of eleven spacetime dimensions. As we don't see these extra dimensions, superstring theory is telling us that *we've so far glimpsed but a meager slice of reality.*

Of course, the lack of observational evidence for extra dimensions might also mean they don't exist and that superstring theory is wrong. However, drawing that conclusion would be extremely hasty. Even decades before superstring theory's discovery, visionary scientists, including Einstein, pondered the idea of spatial dimensions beyond the ones we see, and suggested possibilities for where they might be hiding. String theorists have substantially refined these ideas and have found that extra dimensions might be so tightly crumpled that they're too small for us or any of our existing equipment to see (Chapter 12), or they might be large but invisible to the ways we probe the universe (Chapter 13). Either scenario comes with profound implications. Through their impact on string vibrations, the geometrical shapes of tiny crumpled dimensions might hold answers to some of the most basic questions, like why our universe has stars and planets. And the room provided by large extra space dimensions might allow for something even more remarkable: other, nearby worlds—not nearby in ordinary space, but nearby in the extra dimensions—of which we've so far been completely unaware.

Although a bold idea, the existence of extra dimensions is not just theoretical pie in the sky. It may shortly be testable. If they exist, extra dimensions may lead to spectacular results with the next generation of atom smashers, like the first human synthesis of a microscopic black hole, or the production of a huge variety of new, never before discovered species of particles (Chapter 13). These and other exotic results may